

Embedded Linux & System Software Interview Handbook

Phase 3 – Linux Filesystems & Storage

This phase focuses on how Linux stores, retrieves and manages persistent data. It is essential for kernel developers, BSP engineers and storage subsystem developers.

Chapter 10 – Linux Filesystems

Topics Covered

- Introduction to Linux VFS (Virtual File System)
- Inode, Dentry and Superblock
- File Object and File Operations
- Path Resolution
- ext2, ext3 and ext4 Architecture
- XFS and Btrfs Overview
- procfs
- sysfs
- tmpfs
- debugfs
- ramfs
- OverlayFS
- NFS and Network File Systems
- Flash Memory Basics
- MTD Subsystem
- JFFS2
- YAFFS2
- UBIFS
- UBI Layer
- eMMC, SD Card and NVMe Storage
- Block Layer Architecture
- I/O Scheduler
- Buffer Cache vs Page Cache
- Filesystem Mount Process
- Journaling
- Filesystem Recovery (fsck)
- Storage Performance Optimization
- Filesystem Debugging

- Real Driver Examples
- Senior Interview Questions

Learning Outcomes

After completing this phase you will understand how the Linux Virtual File System abstracts different filesystems, how flash-aware filesystems differ from disk filesystems, how block devices are managed, and how to debug storage and filesystem issues in embedded Linux products.

Phase	Coverage
Phase 1	Computer Architecture to Kernel Synchronization
Phase 2	Memory & Process Management
Phase 3	Filesystems & Storage
Estimated Size	180–250 pages when fully expanded
Target	Senior Embedded Linux / Kernel / BSP Interviews